AgriRisk

AI-Powered Crop Risk Advisory for Smallholder Farmers

Track	Track 3 — Development
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Project ID	

Section 1: Problem Definition & Strategic Alignment

1.1 Problem Statement

Smallholder farmers across Zimbabwe's agricultural districts make critical seasonal decisions input purchase timing, irrigation prioritization, pest response with limited access to consolidated climate and market signal data. Extension officer coverage is thin relative to the farmer population, and existing advisory channels are largely reactive rather than predictive. A farmer who could be warned of elevated risk one week in advance, in a channel they already use, can act before a loss occurs rather than report one after the fact.

1.2 Target Users & Beneficiaries

- Primary Users: Smallholder farmers in Zimbabwe's agricultural districts, reached via a low-bandwidth web-chat interface optimized for limited data connectivity, with a documented road-map toward USSD delivery for farmers without smartphone or data access.
- Secondary Users: Extension officers, agricultural planners, and district-level decision-makers, served by a district-level descriptive analytics dashboard distinct from the core farmer-facing prediction tool.

1.3 Strategic Alignment with Zimbabwe's National AI Strategy (2026–2030)

The National AI Strategy is organized around six pillars: AI talent and capacity development; national AI infrastructure and computational sovereignty; AI adoption and service transformation; governance, ethics and regulation; research, development and innovation; and international collaboration. The strategy names agriculture explicitly as a priority sector, stating that agriculture the backbone of Zimbabwe's economy will leverage AI to boost productivity, strengthen food security, and build climate resilience for smallholder and commercial farmers alike.

This submission is a direct, narrowly-scoped instance of that stated priority a climate-and-market risk classifier delivered through a connectivity-appropriate interface rather than a general-purpose tool retrofitted with strategy language. AgriRisk aligns most directly with two pillars:

- AI Adoption and Service Transformation: the core product a practical, connectivity-appropriate climate-and-market risk classifier delivered to the field.
- Governance, Ethics and Regulation: the data-minimization practices and explicit consent mechanism detailed in Section 4, aligned with the Data Protection Act.

It does not claim alignment with the infrastructure-sovereignty or research-and-innovation pillars at this stage, and the compute roadmap in Section 3 reflects that honestly rather than overstating current technical maturity.

Section 2: Technical Design & Product Logic

2.1 System Architecture & Data Flows

Github link :https://github.com/WilsonNedanhe/agri-ais
The system uses a lightweight, decoupled architecture to serve real-time requests with sub-10ms model inference:

Farmer (web chat widget) → Chat Assistant UI (HTML/JS, consent-gated) → FastAPI Backend [POST /predict · GET /districts · GET /health] → Binary GBT Classifier (sklearn Pipeline) / Descriptive Analytics (district aggregates, no model call) / AI4I sample dataset (synthetic, 360 rows). A separate District Overview view serves the extension-officer / analyst user.

All data flows are verified via an automated pytest suite using the FastAPI TestClient.

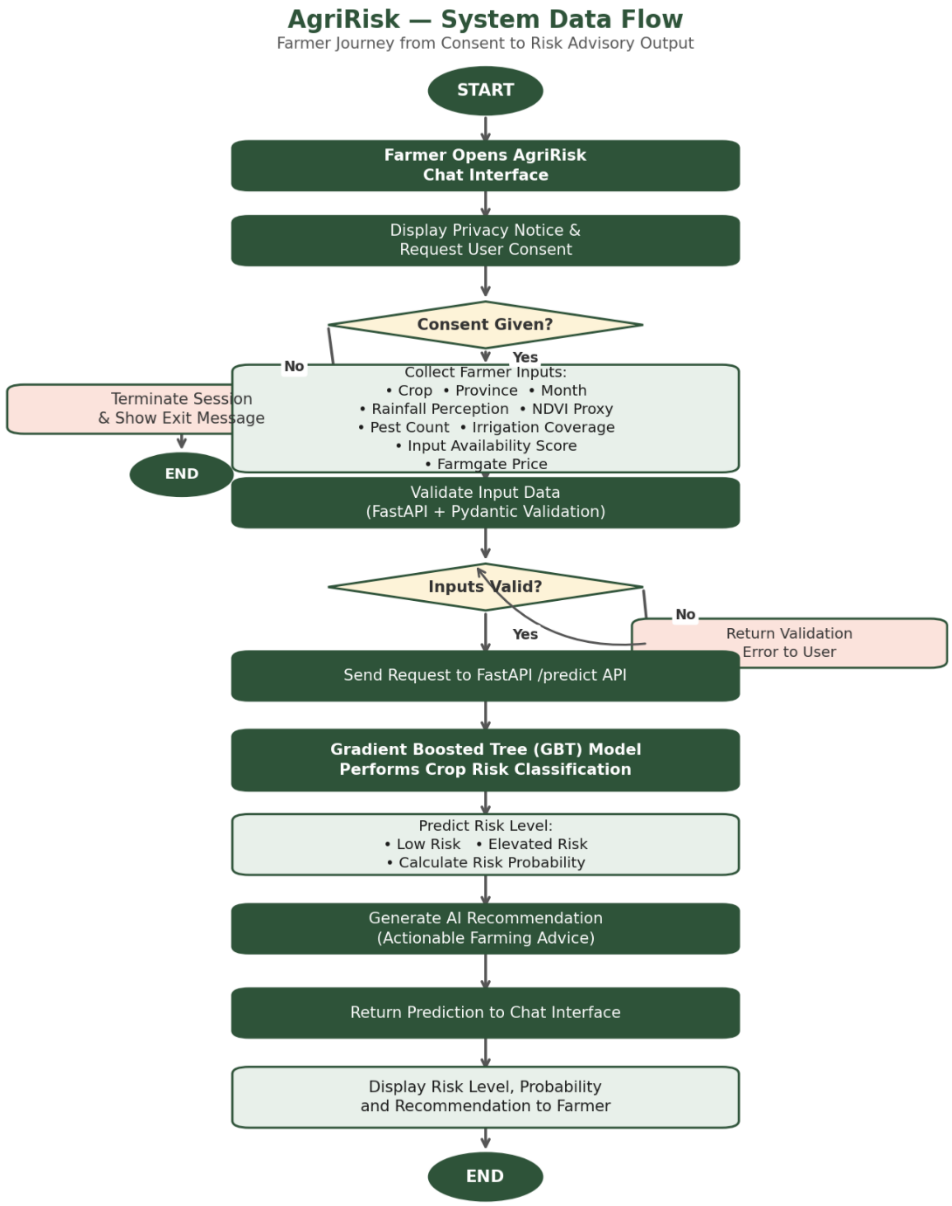
AgriRisk — System Data Flow
Farmer Journey from Consent to Risk Advisory Output
START
Farmer Opens AgriRisk
Chat Interface
Display Privacy Notice &
Request User Consent
Consent Given?
No
Yes
Terminate Session
& Show Exit Message
END
Collect Farmer Inputs:
• Crop • Province • Month
• Rainfall Perception • NDVI Proxy
• Pest Count • Irrigation Coverage
• Input Availability Score
• Farmgate Price
Validate Input Data
(FastAPI + Pydantic Validation)
Inputs Valid?
No
Yes
Return Validation
Error to User
Send Request to FastAPI /predict API
Gradient Boosted Tree (GBT) Model
Performs Crop Risk Classification
Predict Risk Level:
• Low Risk • Elevated Risk
• Calculate Risk Probability
Generate AI Recommendation
(Actionable Farming Advice)
Return Prediction to Chat Interface
Display Risk Level, Probability
and Recommendation to Farmer
END

Figure 1: AgriRisk System Data Flow — Farmer journey from consent to risk advisory output

2.2 Model Architecture & AI Justification

The core predictive engine is a Gradient Boosted Tree classifier (scikit-learn GradientBoostingClassifier, 200 estimators, max depth 4) predicting a binary risk label (Low / Elevated) plus an actionable one-line directive, from nine raw, farmer-answerable inputs: rainfall perception, an NDVI vegetation proxy, pest incident count, irrigation coverage, an input-availability score, farmgate price, month, crop, and province.

- Class Imbalance Handling: balanced sample weighting corrects for the class split (52% Elevated / 48% Low).
- Validation Performance: 5-fold cross-validated balanced accuracy of 0.804 ± 0.040; held-out test precision and recall are 0.79–0.82 across both classes.
- AI Necessity & Anti-Overengineering Audit: a single-variable rainfall threshold baseline scores 75.0% accuracy (74.9% balanced) on the same data. The aggregate gain from the classifier is modest (~6 points), but the more relevant evidence is case-level: in the 40 records (11% of the dataset) where rainfall alone would have told a farmer conditions were safe, the actual outcome was Elevated risk, and the classifier correctly caught 92% of those cases risk driven by compounding pest pressure or poor irrigation access that a single-variable rule structurally cannot represent.
- Dropped Capabilities: a yield-regression module was scoped and deliberately dropped. An initial version scored $R^2 = 0.976$, but a feature-importance audit showed 92% of that explanatory power came from a single crop-identity flag (tomatoes ≈14 t/ha vs. ≈1–3 t/ha for other crops) rather than any climate signal a crop lookup table, not a climate-risk model. It is excluded from this submission rather than presented as an AI capability it does not represent.

2.3 API Layer Engineering

The backend is built using FastAPI and exposes three validated endpoints:

- POST /predict — classification from raw, user-answerable inputs; no derived or composite score is required as input, avoiding a circular dependency present in an earlier draft.
- GET /districts — descriptive district-level statistics (average risk score, percentage of records classified Elevated, dominant crop's average yield) computed from recorded data, without triggering a model inference call.
- GET /health — publicly serves current model metrics and the mandatory dataset provenance / disclosure statement.

All three endpoints are covered by an automated pytest suite (8 tests, all passing at time of submission).

2.4 Production Database Schema

The current MVP reads statelessly from a local, memory-mapped CSV file, with no persistent database. For a production pilot, a minimal Supabase/PostgreSQL schema is proposed, logging anonymized data points only consistent with the data-minimization approach in Section 4:

```
CREATE TABLE predictions (     id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
timestamp TIMESTAMPTZ DEFAULT now(),     crop VARCHAR(50) NOT NULL,     province
VARCHAR(50) NOT NULL,     month SMALLINT NOT NULL,     rainfall_perception INT,
ndvi_proxy NUMERIC(4,3),     pest_count INT,     irrigation_coverage INT,
input_score INT,     farmgate_price NUMERIC(10,2),     predicted_risk_level
VARCHAR(20) NOT NULL,     risk_probability NUMERIC(4,3) NOT NULL,     consent_given
BOOLEAN NOT NULL CHECK (consent_given = TRUE) );
```

No columns exist for names, phone numbers, or hardware identifiers. A month column has been added to the schema so that every model input feature is represented; the field was omitted from the previous draft's schema in error.

2.5 Offline & Edge Device Considerations

This submission does not claim an edge or embedded AI deployment. The chat interface requires an active backend connection; a USSD/feature-phone fallback for non-connectivity users is scoped as a Phase 2 roadmap item (see Section 3), not a July 2026 deliverable. The trained classifier itself is lightweight (a single-digit-kilobyte pickle file, sub-10ms inference measured via the test suite) and would need no meaningful memory or power budget if an edge deployment were pursued later stated as a favourable property for future work, not a built capability.

Section 3: Deliverables & ZCHPC CCE Implementation Roadmap

3.1 Build Lifecycle & Deliverable Milestone Status

Milestone / Deliverable	Target User Segment	Status
Binary GBT risk classifier trained, cross-validated, feature importances checked	System Kernel	Complete
FastAPI backend (/predict, /districts, /health), pytest-verified	Application Layer	Complete
Farmer chat widget (consent-gated, qualitative-input mapping, demo-mode fallback)	Smallholder Farmers	Complete

District Overview metrics dashboard	Extension Officers / Planners	Complete
Formal Track 3 technical proposal document	Adjudication Board	In Progress
Real, AGRITEX-sourced problem statistics for Section 1	Zimbabwe Districts	Outstanding
USSD channel integration via AfricaTalking sandbox	Offline / Feature-phone Users	Phase 2
ZCHPC CCE deployment and parallelised hyperparameter sweep	AI Infrastructure	Phase 2

3.2 Compute Requirements & Resource Footprint

Because AgriRisk relies on classical machine learning rather than a deep learning architecture, its resource profile is minimal and should be described as such rather than inflated to match a larger compute narrative:

- Training Footprint: under 60 seconds on a standard commodity laptop CPU; no GPU required.
- Inference Latency: sub-10ms per prediction, measured via the test suite.
- Production Hosting: supported by a single low-cost VPS or free-tier instance.

3.3 Dependency-Locked Manifest (requirements.txt)

```
fastapi==0.139.0   uvicorn==0.51.0   scikit-learn==1.8.0   pandas==3.0.2
pydantic==2.13.4   pytest==9.1.1
```

3.4 ZCHPC CCE Plan

No ZCHPC deployment has been performed as part of this submission. The stated plan for the CCE implementation phase is:

- Scale Training Operations: retrain the GBT pipeline once larger, verified agricultural datasets are consolidated from AGRITEX, FAO, and CIMMYT sources, at a scale beyond the current 360-row sample.
- Parallelized Hyperparameter Optimization: replace the current manually-chosen parameters with a ZCHPC-backed GridSearchCV sweep over extended ranges of n-estimators, max_depth, and learning rate.
- Concurrency Load Testing: benchmark FastAPI inference latencies (p50, p95, p99) under simulated multi-channel traffic to confirm runtime stability before piloting.

This is stated as intent for the post-Challenge development phase, not a completed activity.

Section 4: Compliance & Risk Mitigation

4.1 Ethical Safeguards & Explanations

The interface discloses when it is running in demo mode (client-side simulated logic) versus returning a live, backend-verified model prediction, rather than presenting both identically. The public GET /health endpoint, and this document, disclose the synthetic nature of the 360-row training dataset and the deliberate exclusion of the unreliable yield-prediction module, rather than allowing an inflated capability claim to stand.

4.2 Data Protection Act [Chapter 12:07] Compliance Framework

AgriRisk enforces a structural data-minimization framework. No personally identifying data — names, national ID numbers, mobile numbers, or exact coordinates is collected, logged, or processed. Inputs are limited strictly to anonymous agronomic and environmental signals necessary for the prediction task.

- Consent Mechanics: because the interaction channel is a chat interface rather than a form, consent is captured as an explicit, mandatory user action ("Yes, I agree") at the start of every session, logged with a timestamp, rather than assumed. No session proceeds without this step.
- Stateless Logging: because current interactions are entirely stateless, no persistent farmer identifier is stored, so there is no personal record to erase on request in the current MVP. If a persistent log database is introduced in production (see Section 2.4's proposed schema), a verified right-to-forget deletion API endpoint must be implemented before that change ships.

4.3 Model Unit Testing Matrix

An automated pytest suite (8 tests, all passing at time of submission) validates:

- Correct response schema and probability normalization on /predict.
- HTTP 422 rejection of out-of-range inputs (e.g., NDVI outside 0.0–1.0) and missing required fields, rather than silent bad predictions.
- Graceful handling of an unseen, out-of-vocabulary crop value without a server crash.
- Correct row count and province-filtering (geographic isolation) behaviour on /districts.
- Verification that /health's dataset statement discloses both the synthetic-data limitation and the dropped yield module.

4.4 Cybersecurity Hardening & Disclosed Gaps

Input Sanitation: all incoming request vectors are parsed and validated against strict Pydantic type, range, and required-field schemas before reaching the model, rejecting malformed requests structurally.

Disclosed Gaps (MVP), listed rather than hidden:

- The current CORS policy (allow_origins=["*"]) is permissive and intended for demonstration only; it must be restricted to specific production domains before public deployment.
- No authentication (API key or token) is currently implemented on the backend — acceptable for a Challenge demo, not for production use.
- No rate limiting is currently implemented, leaving the endpoint open to abuse at scale.
- TLS/HTTPS termination is expected to be handled by an upstream reverse proxy or hosting platform and has not yet been configured.

Section 5: Sustainability & Future Adoption

5.1 Expected Operating & Business Model

AgriRisk is structured as a public-service tool rather than a direct-to-farmer monetisation product. Long-term operational sustainability relies on partnership with AGRITEX or a district agricultural cooperative, providing both high-fidelity field-data access and distribution credibility among farming communities.

To bridge the connectivity divide, the model will leverage the lead innovator's existing telecommunications development work within the AfricaTalking USSD sandbox environment, creating a route to reach farmers without smartphone access at effectively zero marginal cost per session — once that channel is built out in Phase 2.

5.2 Financial Cost Projections

Item	Estimate
Core backend hosting	\$5–\$20 / month at MVP scale (small VPS or free-tier cloud instance)
Model retraining / compute	Negligible — CPU-only, under 1 minute per run
USSD infrastructure access (Phase 2)	To be determined, based on session-volume pricing arranged with a telecom aggregator
Real dataset sourcing (AGRITEX / FAO / CIMMYT)	To be scoped once a formal data-sharing partnership is confirmed

5.3 Licensing

The current AI4I sample dataset carries no intellectual property or licensing restriction for Challenge use. Any real production data introduced via AGRITEX, FAO, or CIMMYT will carry its own licensing terms.

5.4 Scaling and Sustainability Pathway

The immediate next step following the Challenge is a pilot partnership with a single designated AGRITEX district office, evaluating model predictions against real, physical end-of-season yield outcomes before any broader claim of effectiveness is made.

Zimbabwe's National AI Strategy references funding mechanisms, including a national AI fund, intended to support exactly this kind of practical, sector-specific AI adoption. This submission's continued development is a plausible fit for that funding route, subject to confirming eligibility criteria once published in detail.